

Development of a RASSOR Rover Remote Operation System with AI-Driven Digital Twin for
Latency Compensation

PROJECT PROPOSAL

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1. Introduction and Case Statement

1.1 Introduction

Space exploration continues to expand our scientific knowledge and technological capabilities, contributing to a deeper understanding of extraterrestrial environments. While long-term goals often include increasing the number of manned missions, human spaceflight requires significant financial investment, complex logistical coordination, and exposes astronauts to considerable risk. For this reason, robotic systems play a critical role in supporting exploration efforts and reducing direct human exposure.

One persistent challenge in space operations is communication latency. Due to the great distance between the Earth and the Moon, there is an inherent delay in transmitting commands and receiving feedback, commonly referred to as lunar latency. Even a delay of only a few seconds can become critical during navigation or obstacle avoidance tasks as operators have to make decisions based on outdated information. Over a short span of time, this lag can significantly impact mission performance, safety, and efficiency.

1.2 Case Statement

To address this operational challenge, this project proposes the development of a remotely operated RASSOR (Regolith Advanced Surface Systems Operations Robot - we could do an acronym page or leave it as this) rover system designed to function under simulated communication delay conditions. This will be a two-semester collaboration between the University of Central Florida (UCF) and the Universidad Politécnica de Durango (UPD) establishing a binational learning and research environment focused on advancing remote rover operations. Each institution will independently construct a RASSOR rover and conduct cross-institution teleoperation through Internet-based communication channels, intentionally introducing artificial time delays to replicate lunar latency.

This approach allows the team to evaluate rover performance under realistic operational constraints while developing technical strategies to mitigate the effects of delayed communication. The system will integrate an RGB-D sensing platform to capture synchronized visual and depth data for environmental mapping and obstacle detection. Building upon this data, a digital twin framework will be developed to estimate the rover's predicted state during communication gaps, enhancing operator awareness and improving navigation reliability.

The Florida Space Institute (FSI), serving as the project's client and technical advisor, seeks to advance rover system capabilities and explore innovative methods for improving remote robotic performance in space exploration contexts. Current teleoperation systems primarily rely on delayed sensor feedback, limiting the operator's ability to react proactively to dynamic terrain conditions. Without predictive support mechanisms, latency increases the risk of collisions, inefficient path course, and potential hardware damage. These risks directly affect operational cost, system longevity, and overall mission effectiveness.

By implementing simulated communication delays within a controlled testing environment, this project establishes a structured framework for evaluating rover performance under latency constraints. The integration of RGB-D sensing and a digital twin platform enables forward-state estimation and improved situational awareness during teleoperation. Through this approach, the project provides FSI with a validated prototype

system and documented methodology that can support continued development, optimization, and future research initiatives.

2. Problem Statement

2.1 Definition of the Problem

Within the context of lunar and planetary exploration, primary challenges tend not to lie in the rover mobility itself. The issues begin to arrive in the delayed interactions between the human operator and the rover. Current remote rover operations suffer from: communication latency, limited environmental awareness, and a lack of synchronized feedback systems. In turn, this reduces operational efficiency, rover safety during remote navigation, and the discovery of beneficial findings that advance exploration. However, communication latency between Earth and the Moon introduces an approximate three-second one-way delay.

Real sensor data received on Earth represents the rover's past state rather than its current condition. The six-second round-trip delay disrupts the control loop between the human operator and the rover. This forces decisions to be made using outdated information, which in turn leads to navigational accuracy decreasing as well as operational efficiency. In lunar and planetary exploration, where precision and timing are critical, this disconnect can significantly impact the rover's safety and the performance of the mission.

2.2 Causal Factors

The communication delay arises from the physical distance between the Earth and the Moon, which creates unavoidable signal latency as well as how the communication arrives. The communication pathway for lunar rovers is as follows:

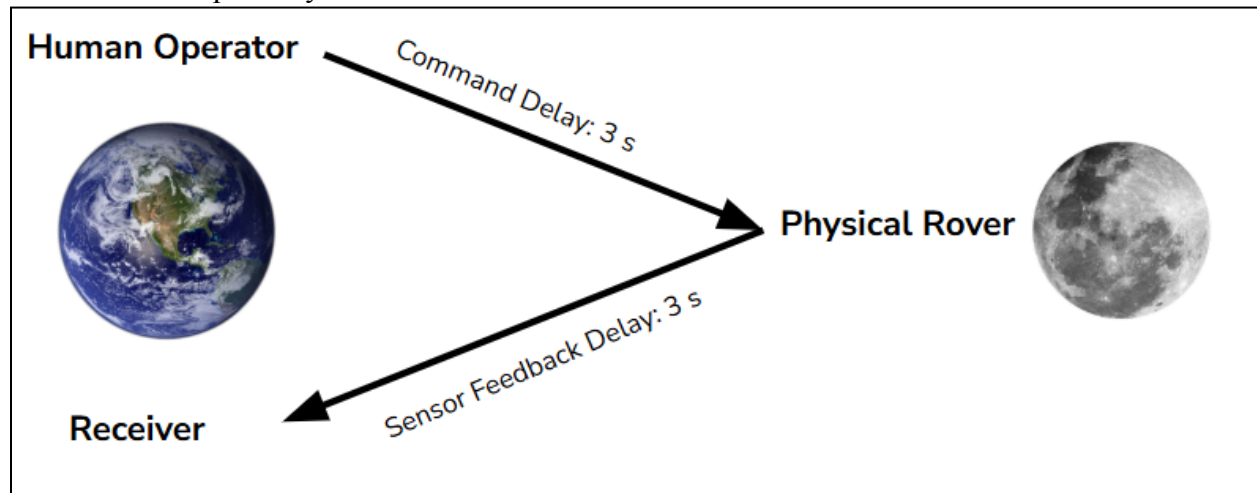


Figure 1: Earth to Moon delay visualization

The human operator sends commands out to the Deep Space Network (DSN - a global network of large radio antennas). The command is transmitted as a radio signal that travels at the speed of light. The lunar orbiter or the surface rover receives it, and then the command is executed. Afterwards, the rover sends the data back through the same chain. Beyond the technicality of how the transmission operates, other causal factors that are consequently affected by latency are technical system designs, network infrastructure, as well as human factors.

Traditional teleoperation systems assume near real-time feedback, which cannot be replicated under Earth-Moon communication conditions. A teleoperation system consists of a telemanipulator that a human operator controls at a distance and one or more robotic devices that telemanipulator can control (Linnell, 2024). However, the architecture of teleoperation systems

is not designed to sustain communication latency. Since these systems are structured for continuous, real-time adjustment, the lack of an integrated predictive state built into the rover systems to manage this delay can cause command and feedback to be desynchronized. Latency amplifies inaccuracies rather than being mitigated within the design of the system.

Constant delay is manageable within a control system; however, variable delay is destabilizing, which is far more difficult to manage. Network infrastructures are designed in a manner in which they are subject to network variability, which makes delays inconsistent. Rather than experiencing a fixed three-second delay, communication may fluctuate by milliseconds around that baseline, which introduces uncertainty. Network latency is the delay in data communication over a network, packet loss is when data packets fail to reach their destination, network jitter refers to the variability in packet delay at the receiver (Freedman, 2026).

Understanding the interconnection between these three variables explains why the physical latency between Earth-Moon communication is not the sole contributor to the arrival of data from the source. As data travels through these processes, it is affected by traffic congestion, signal interference, and bandwidth limitations. Together, these infrastructure-related variables compound the latency effect and can create unpredictable decision-making, further destabilizing remote rover control.

Human operators rely on consistent feedback to maintain situational awareness and perform precise control adjustments. In teleoperation environments, visual and sensor feedback enables operators to make adjustments to steering and obstacle avoidance of the rover. Communication latency directly leads to performance degradation as decisions are forced to be made using outdated information. In turn, this reduces overall system performance as well as navigational errors. Over time, these margins can compromise mission objectives, which limit the venture of further lunar and planetary missions.

2.3 Organizational Impact

Communication latency reduces the operator's ability to respond to environmental hazards such as craters, steep slopes, and uneven terrain. Lunar rovers operate within constrained environmental conditions that limit their ability to explore. Since most rovers rely on solar power, they are limited to operational windows within the fourteen-day lunar period. During the fourteen-day lunar night cycle, they either hibernate or rely on battery power. Misnavigation caused by delayed control response may result in the rover being stranded. During the lunar night, rovers face extreme temperatures, solar and cosmic radiation, and lunar dust, which can degrade the solar panels, clog mechanisms, and potentially render the rover inoperable. Operational failure under these conditions impacts mission reliability, organizational objectives, as well as continuous research.

2.4 Significance of Addressing the Problem

The absence of a predictive state within traditional teleoperation systems highlights the necessity of an AI-driven digital twin. A digital twin provides a virtual representation of the rover that mirrors incoming visuals and predicts the rover's short-term position during the round-trip time delay. Rather than eliminating the latency issue entirely, the digital twin aims to reduce the operational impact that stems from it. The digital twin is expected to mirror the rover's actual state on the lunar surface and perform corrective actions.

The integration of a RGB-D camera, attached to the rover, will help with data acquisition that will be beneficial to the digital twin, being able to make positive predictive corrections. By leveraging machine learning models with the data received by the camera, it enhances

environmental awareness and predictive modeling. The combination of real-time sensor data and trained AI models will allow the digital twin to mitigate the dangers presented. The digital twin will improve situational awareness by reducing the likelihood of navigational errors under delayed conditions by telehuman operators within the round-trip time delay.

By addressing latency through predictive modeling, the results from the digital twin via intelligent teleoperations improve rover safety, operational efficiency, and mission reliability. Beyond that, they allow improved data acquisitions of exploratory missions. Which in turn, provide accelerated, efficient, and successful operations. This creates a framework for future deep-space missions where latency is even greater than those experienced in Earth-Moon scenarios.

3. Goal Statement and Project Objectives

The goal of this project is to improve the operational performance of a remotely operated RASSOR rover system by addressing the disruption in the human-rover control loop caused by Earth-to-Moon communication latency. As described in sections 2.1-2.4, latency forces operators to make decisions based on outdated state information, which increases navigation error, destabilizes control, and reduces mission reliability.

To tackle in-field communication latency, this project integrates a predictive digital twin capable of estimating the rover's current state during communication delay. The digital twin will be informed by real-time environmental and positional data captured through an OAK-D Lite RGB-D sensing system. This sensor enables synchronized color and depth data acquisition, allowing the system to collect spatial mapping information and rover motion data necessary to train and refine the predictive model.

By analyzing the teleoperation process and incorporating sensor-driven state prediction, the project aims to improve navigation accuracy, enhance operator situational awareness, and increase overall mission performance under simulated lunar latency conditions. The anticipated organizational benefit for the Florida Space Institute (FSI) is improved rover safety, reduced operational risk, and enhanced reliability of remote robotic exploration systems operating in latency-constrained environments.

3.1 Project Objectives

- To demonstrate reliable rover operation using manual-only rover operation and a predictive digital twin system under approximately 3-second simulated Earth-to-Moon communication latency by April 15th, 2026
- To reduce obstacle collision frequency, as measured through controlled trial comparisons under latency conditions, between manual-only rover operation, digital twin-assisted operation, and AI-implemented digital twin operation by April 15th, 2026
- To reduce the average obstacle course completion time under latency conditions between the manual-only rover operation, digital-twin assisted operation, and AI-implemented digital twin operation by April 15th, 2026
- To reduce average path deviation from the predefined optimal route under latency conditions using digital-twin assisted operation, compared to manual-only rover operation, and AI-implemented digital twin operation by April 15th, 2026

4. Project Scope and Constraints

4.1 Project Scope

Throughout the course of this project, the team will

- Construct one functional RASSOR rover
- Setup an internet based teleoperation control of the rover

- Establish Artificial communication delays in the rover communication
- Integrate an RGB-D sensing system
- Develop a digital twin framework to estimate rover state during communication gaps
- Collect and analyze performance metrics
- Document system architecture, testing procedures, and results
- Provide recommendations for continuation of the project in the second semester

4.2 Outside Project Scope

Throughout the course of this project the team will not

- Develop systems in actual lunar or space environments
- Utilize space qualified hardware and systems
- Fabricate custom hardware outside the rover platform and sensor mounts
- Conduct durability testing on the rover
- Develop satellite communication infrastructure

4.3 Project Assumptions

- Internet Connectivity is sufficiently stable for teleoperation experiments
- Operators will have a sufficient baseline skill prior to testing rover operation
- Indoor or controlled outdoor test environments reasonably approximate key rover navigation challenges
- RGB-D sensors provide sufficient accuracy for obstacle detection and mapping
- An artificial approximate three second delay adequately represents lunar delay for testing purposes

5. Project Timeline

The project begins with planning activities such as the kickoff meeting, materials list and budget submission, and development of the project plan, culminating in the Proposal Presentation and Final Proposal Report submissions in mid-February. Following proposal approval, the schedule transitions into hardware development, including base and wheel construction, electronic component planning and implementation, and three stages of manual drive testing to validate rover mobility. Digital Twin development begins in early March and overlaps strategically with hardware testing to allow iterative refinement. AI integration follows completion of the digital twin baseline, leading into combined system testing in early April.

In parallel with technical development, structured reporting milestones are embedded throughout the timeline, including the AS-IS Presentation and Report in late March and the To-Be Report, Showcase Video, and Finalist Face-to-Face presentation in April. The chart demonstrates coordinated parallel workstreams and logical task sequencing, supporting efficient integration of mechanical, electrical, software, and analytical components. Overall, the Gantt chart reflects a realistic, milestone-driven execution plan designed to deliver a functional rover prototype, an established digital twin framework, and documented results within the academic semester.

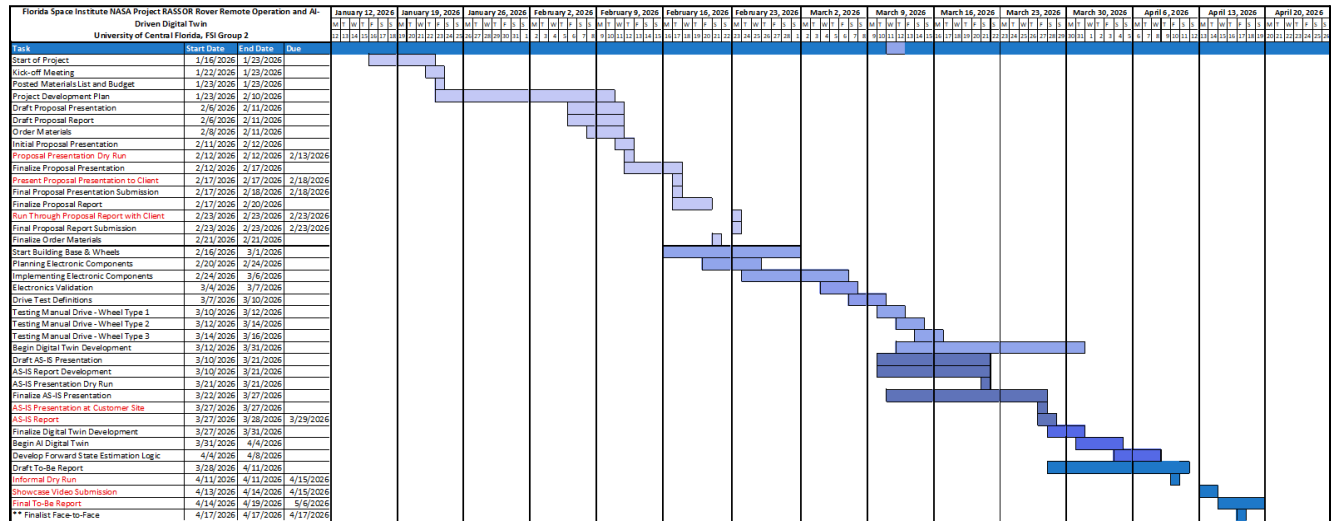


Figure 2: GANTT Chart of Project

6. Proposed Methods

For this project, we will apply Industrial Engineering principles to approach the communication latency challenge from a systems perspective rather than just a technical one. We will define stakeholder requirements and break the rover system into functional components to ensure that mobility, sensing, communication, and predictive elements are clearly structured and properly integrated. Using process mapping techniques, we will analyze the full teleoperation workflow, including command transmission, sensor feedback, and delay propagation, to identify where latency creates inefficiencies or increases operational risk. Because this system relies heavily on human-in-the-loop control, we will also incorporate human factors engineering to evaluate operator workload, situational awareness, and interface clarity under delayed feedback conditions. To validate improvements, we will establish measurable performance metrics such as task completion time, navigation deviation, collision frequency, and response delay, and analyze system performance through controlled testing scenarios. Additionally, structured risk assessment tools will be used to proactively identify potential technical and operational risks that could impact reliability or cost. Finally, we will apply project management and documentation practices to ensure continuity between semesters and provide a clear foundation for the team that will continue development. By using an Industrial Engineering framework, we ensure that latency is addressed as a system-wide design challenge, emphasizing integration, measurable performance improvement, and long-term sustainability.

While the main focus of this project relies on principles from Industrial Engineering the team will utilize Data Science principles in the latter section of the project. Data acquisition will be set up through a camera on the rover to capture information on how it moves within the environment. Data from these sensors enables reinforcement learning to improve performance of the predictive model in analyzing motion of the rover. For future analysis, methods of machine learning will be considered, primarily a neural network, for predicting how commands will affect the motion of the rover, and determining the pathfinding going forward.

7. Standards

The fabrication of rover components using additive manufacturing will adhere to 3D printing safety guidance provided by Stanford Environmental Health & Safety. These guidelines

identify potential hazards associated with 3D printing, including emissions of ultrafine particles, volatile organic compounds, and thermal risks from heated components. To mitigate these risks, printing will be conducted in a well-ventilated environment with appropriate operational controls and adherence to manufacturer specifications. Implementing these measures ensures safe production of prototype components while minimizing health and environmental risks.

ACM Code Of Ethics

The proposed system adheres to the ACM Code of Ethics and Professional Conduct for responsible computing practice. The code emphasizes maintaining high standards of professional competence, ethical behavior, and integrity, as well as respecting the intellectual work required to produce new ideas, inventions, creative works, and computing artifacts. Incorporating these principles ensures that the rover framework is developed and implemented in a manner that is responsible, reliable, and consistent with accepted professional and ethical standards.

NASA Material Safety Data Sheet

Testing in simulated lunar soil will follow the NASA Material Safety Data Sheet (MSDS) for JSC-1A lunar regolith simulant used in the Exolith Lab. The guidelines recommend minimizing dust inhalation, washing exposed skin after handling, and using appropriate personal protective equipment to ensure safe operation during rover testing in a simulated lunar environment.

Safety Factors

During the assembly of the rover, usage of PETG filament presents risks that must be considered. Printing operations will strictly follow OSHA standards and also 3D printing safety standards, such as supervised operation and temperature control. Using personal protection gear, such as safety goggles and gloves, will be required when necessary.

After assembly, testing of the robot will most likely be conducted in the simulated moon environment. The regolith moon dust presents inhalation and skin exposure risks. The MSDS provides recommendations for testing in this environment.

Public Health Factors

Public health regulations will be followed during the development of the rover. Proper 3D printing practices, such as using PETG filament and ensuring proper ventilation, will prevent exposure risks to students and any other person working in the FSI lab.

As the exolith lab was designed, created, and maintained to follow safety regulations, testing exclusively in the lab will ensure the project aligns with public health standards.

Ethics and Welfare Factors

This project aligns with NASA's ethics framework by emphasizing responsible artificial intelligence development, transparency, and human oversight. The design of the digital twin ensures that human decision making is still essential for rover operation. Retaining this control structure ensures accountability and reduces ethical risks of autonomous decision making.

Global Factors

As this project is a collaboration between UCF and UPD, global factors are essential to consider. Global communication requires consistent documentation and scheduling. Differences in budget, technology, and university resources will be addressed to foster proper project development.

Space exploration inherently carries global implications, including international cooperation, political considerations, and shared scientific advancement. The framework developed in this project can be applied to other institutions developing similar projects.

Cultural Factors

The project team consists of members from diverse cultural backgrounds. Bringing awareness to this promotes equality and equitable task distribution. The project will be developed with a mutual understanding of equality for all team members.

Other cultural factors are considered in working with UPD. Differences in culture and institutional practices are addressed through clear communication and documentation. Again, the project team's emphasis on equality and mutual respect strengthens international collaboration.

Social Factors

While advancements in robotics and AI influence can improve workplace efficiency and safety, a social factor to consider is labor displacement. The project emphasizes augmentation rather than replacement of human roles. Humans are still essential for decision making and operation of the rover. By integrating this type of human operated design, the project aligns with ethical adoption of robotics and AI.

Environmental Factors

The usage of 3D printers and PETG filament requires responsible material management. All required environmental regulations will be followed during handling of the printer, electrical components, and battery parts.

As the project is tested under simulated moon conditions, it has an environmental advantage over actual interaction with a lunar environment. Also, the digital twin framework allows for more efficient training and testing here on earth.

Economic Factors

The project operates under a constrained university budget. Preliminary spending, as documented below, is well within range of this budget. Only printing required rover components under optimal conditions will reduce filament waste and thus reduce expenditures. Safe and mindful handling of electrical and battery components will also reduce waste and reduce expenditure.

Risk Factors

Technical risks include potential inaccuracy of digital the digital twin. This will be addressed through model validation with real sensor data. Operational risk factors arise through coordination challenges with the UPD. This will be addressed through consistent communication and documentation of project development. Potential cost overruns are a financial risk factor that can occur at any stage of development. However, the highest risk of overrun is during printing due to potential excess filament waste. This will mitigated through careful printing practices.

8. Key Project Deliverables

- A completed, drivable rover - The final rover built and operating, able to be driven, and outfitted with sensors for positional data acquisition.
- Digital Twin System - A physics based Digital Twin of the physical rover that can model the effects of motion commands on the rover.
- Architecture for AI assisted motion control - Setup for autonomous AI rover navigation, to be built upon in the second semester of the project.

- Handoff Documentation - Well organized documentation of the entire process of design and control, to allow the second semester to have the necessary information to take over the project.

9. Project Management Plan

An Integrated Master Schedule (IMS) has been developed and is attached to this report for reference. The schedule provides a detailed breakdown of the project Gantt chart, including itemized tasks, subtasks, defined milestones, and assigned responsible team members for each activity. This structured schedule ensures clear task ownership, accountability, and alignment across hardware development, digital twin implementation, and reporting deliverables throughout the project lifecycle.

10. Project Evaluation Plan

10.1 Formative Evaluation Plan

To ensure the project completes its objectives on time, at minimum weekly meetings will be held with all parties involved. At this time, the current progress on the project, and state of design will be evaluated, and goals will be set for the upcoming week. Through this the project can stay on schedule, ensuring time is allocated adequately to complete the project by end date. Documentation will be taken at each meeting to have a clear reference for what was discussed and what was set to be completed.

10.2 Summarative Evaluation Plan

To determine if the Digital twin and AI assisted rover control have improved rover performance, the finished rover will be put through rounds of testing in obstacle courses. Tested under the simulated delay conditions, the rover will be operated both with and without these augmenting control measures, and performance in completing the course will be measured. Measurements of success may include such measures as time to course completion, number of collisions with obstacles, and amount of deviation from planned paths. Comparing the results of the rover when driven with the digital twin, and without, will determine if the model is successful in improving rover navigation under time delay conditions.

11. Project Cost Estimate

Item	Quantity	Unit Price	Total Cost	Source	Explanation
OAK-D Lite Sensor	1	180	\$ 180.00	Luxonis	Runs the AI model required for rover training and obstacle detection.
Polymaker PETG Yellow Filament	4	19.99	\$ 79.96	Amazon	Used to 3D-print rover parts in durable PETG material.
Polymaker PETG Blue Filament	4	19.99	\$ 79.96	Amazon	Additional PETG filament for printing structural rover components.
Polymaker PETG White Filament	4	16.99	\$ 67.96	Amazon	PETG filament for printing rover body parts.
Polymaker PETG Black Filament	4	16.99	\$ 67.96	Amazon	PETG filament for printing rover body parts.
ELEGOO Filament Dimensional Accuracy Printer	1	51.96	\$ 51.96	Amazon	Ensures accurate printing of rover components.
DISEN Vanadium Steel Allen Wrenches	2	13.99	\$ 27.98	Amazon	Required to install Allen screws and assemble 3D-printed parts.
Libre Computer AML S905X CC V2	2	60	\$ 120.00	Amazon	Acts as the rover's main processing unit to run software.
Arduino Projects Compliant Kit	2	22.99	\$ 45.98	Amazon	Provides microcontroller components needed for rover electronics.
Arduino Control Board	2	9.39	\$ 18.78	Amazon	Controls sensors, motors, and other rover electronics.
Stepper Motor Driver	2	10.19	\$ 20.38	Amazon	Drives the stepper motors that move the rover.
Nano Wireless Network Dongle	2	10.01	\$ 20.02	Amazon	Provides wireless communication for rover operation.
SanDisk Ultra 32GB Memory Card	2	17.8	\$ 35.60	Amazon	Stores rover software, logs, and AI model data.
STEPPERONLINE Stepper Cable & Connector	8	14.99	\$ 119.92	Amazon	Connects stepper motors to the control system.
LiFePO4 Deep Cycle Battery	1	55.98	\$ 55.98	Amazon	Powers the rover during operation.
Battery Charger	1	39.99	\$ 39.99	Amazon	Recharges the rover's battery.
Emergency Stop Switch	2	11.99	\$ 23.98	Amazon	Allows forced shutdown of rover movement for safety.
Arducam Camera	2	34.99	\$ 69.98	Amazon	Provides visual input for navigation and monitoring.
Buck Converter	2	7.98	\$ 15.96	Amazon	Regulates voltage for rover electronics.
Power Splitter Cable	2	5.8	\$ 11.60	Amazon	Splits power to multiple camera components.
Type-C Male Plug Pigtail Cable	2	8.99	\$ 17.98	Amazon	Connects power and data to components requiring USB-C.
Alloy Steel Screws	2	19.68	\$ 39.36	Amazon	Secures 3D-printed rover parts together.
SD Card Reader	2	11.99	\$ 23.98	Amazon	Allows reading/writing rover SD cards on a computer.
Plug Connector Cables	2	9.89	\$ 19.78	Amazon	Connects electrical components throughout the rover.
Electrical Wire Cable	1	18.99	\$ 18.99	Amazon	Provides wiring for power and signal connections.
Spade Connectors	1	14.99	\$ 14.99	Amazon	Helps connect wiring securely.
Shrink Fork Connectors	1	9.99	\$ 9.99	Amazon	Provides insulated, durable wiring connections.
Cooling Fans	2	9.99	\$ 19.98	Amazon	Prevents rover electronics from overheating.
Cable Cutter	2	8.99	\$ 17.98	Amazon	Cuts wiring during assembly.
Socket DIY Cord	2	11.15	\$ 22.30	Amazon	Provides additional wiring and power routing.
Steel Bearings	4	7.19	\$ 28.76	Amazon	Supports mechanical movement of rover components.
Deep Groove Ball Bearings	2	9.99	\$ 19.98	Amazon	Ensures smooth wheel and joint rotation.
Screw Set	2	8.99	\$ 17.98	Amazon	Provides assorted screws for assembly.
Hexagon Nuts	1	5.99	\$ 5.99	Amazon	Secures screws in rover frame.
M2 x 30 mm Socket Cap Screws	1	9.99	\$ 9.99	Amazon	Used for fastening specific rover components.
M4 x 0.7 mm Nylon Lock Nuts	1	5.99	\$ 5.99	Amazon	Prevents screws from loosening due to vibration.
M4 x 12 mm Alloy Steel Hex Screws	2	6.49	\$ 12.98	Amazon	Additional fasteners for rover assembly.

Table 1: Itemized list of parts and their prices

The project cost will be used towards the rover whether it is parts for the rover or help in training the AI for the rover. As shown in the chart above is the explanation of why each part is needed in order to make a functional rover with AI. PETG filament is used to 3d print the rover. The wrenches, compliant kit, cable cutter and screw sets are tools used for the team to help assemble the wiring and 3d printed pieces. The control board, motor driver, network dongle, memory card, stepper cable, battery, stop switch, camera, cables, screws, connectors, fans, bearings, and nuts all will be bought and then installed on the rover. The OAK-D lite sensor will be bought through the vendor's website (Luxonis, n.d.-b). The rest of the items will be purchased through Amazon. The grand total will cost \$1461.55.

12. Risk Management (Project)

Risk Assessment Matrix						
Likelihood	Very High			6		1
	High					
	Medium	4	7	5		2
	Low			3		
	Very Low	8				
		Very Low	Low	Medium	High	Very High
		Impact				

Risk	
1	Communication Latency
2	Rover Hardware Failure
3	Camera Lead Time Delay
4	3D Printer Delivery Delay
5	AI Digital Twin Inaccuracy
6	Integration Issues between AI and Rover
7	Rover Control Compatibility with UNIPOLI
8	Environmental Risks

Figure 3: Risk Assessment Matrix

The Primary risks to the project relate to communication latency, hardware reliability, AI accuracy, Integration, Supply Chain delays, Cross functional coordination and environmental factors. Risk levels are determined based on Likelihood and impact. Mitigation strategies are implemented to reduce the probability of occurrence or the severity of the consequences.

Communication Latency is a primary risk resulting from the simulated delay conditions. There is a high likelihood that operators will experience delayed feedback, which may reduce real-time responsiveness and increase the risk of incorrect maneuvering. The impact of this risk is high because it directly affects rover control and mission performance. To mitigate this risk, the team will implement AI-assisted prediction through a digital twin model that simulates rover motion ahead of the physical execution. This will allow the team to be proactive when making decisions despite the latency.

Rover Hardware failure can be a significant risk because of potential network instability or hardware malfunction during teleoperation. The likelihood of this is moderate but the impact is high because hardware failure could slow testing, damage equipment, or interrupt the presentation from the different universities. This risk might come from communication interruptions, electrical issues or mechanical component stress. To mitigate this risk, a rover failsafe safe stop mode will be implemented to immediately pause motion and stabilize the system if a fault or communication loss is detected.

A risk exists regarding delays in camera component delivery due to late arrival of parts. The likelihood of this occurring is moderate, and the impact is medium because delayed camera integrations might postpone testing of visual feedback and AI training activities. This risk mostly would come from supply chain variability. Mitigation includes early preparation and procurement of rover components to make sure integration activities can continue without disrupting the schedule.

Another potential risk is associated with the 3D printer delays caused by equipment shipment issues. The likelihood of this risk is low but the impact is medium since rover fabrication and replacement parts depend on timely printing capabilities. To reduce this risk, the team plans to utilize alternative 3D printing resources if the primary equipment is unavailable.

AI digital twin inaccuracy is a moderate risk because of limited data training during the early phases. If the model inaccurately predicts rover behavior or terrain interaction, operator decision making could be negatively affected. The impact is medium because inaccuracies may reduce system reliability but are unlikely to cause major failure due to human supervision. Mitigation includes maintaining human oversight and continuously refining the AI model with real sensor data to improve prediction accuracy.

Integration issues between AI and rover can present a moderate risk with medium impact. Software compatibility issues, data formatting inconsistencies, or control interface mismatches may show up during system integration. These issues could delay testing and reduce overall system efficiency. To mitigate this risk, the team will conduct incremental testing throughout development, validating individual subsystems before full integration.

Because this project involves cross-institution collaboration, differences in rover configuration may create compatibility issues. The probability of this is moderate, and the impact is medium since this could delay joint teleoperation demonstrations. This risk comes from configuration differences and coordination challenges. Mitigation will involve maintaining open and consistent communication between teams and potentially joint testing sessions.

Environmental risks such as unfavorable weather conditions affecting outdoor testing, are considered low likelihood and low impact due to the availability of controlled indoor facilities. Testing conducted in the Exolith lab environment reduces exposure to environmental variability so this means the mitigation for the risk is to prioritize indoor testing and scheduling flexibility.

13. Summary and Conclusions

Communication latency remains one of the most critical operational barriers in lunar and planetary rover teleoperation. The inherent three-second delay in transmission between Earth and the Moon disrupts real-time control, weakens situational awareness, and introduces safety and efficiency risks. Traditional teleoperation architectures, designed for near-instantaneous feedback, are not structured to manage persistent or variable delay. As a result, rover systems remain vulnerable to navigational inaccuracies, hardware damage, and mission inefficiencies from this gap, which can lead to full operational failure.

The proposed development of a remotely operated RASSOR rover system, integrated with RGB-D sensing and a predictive digital twin framework, directly addresses these limitations. By simulating lunar latency in a controlled test environment, the project establishes a realistic platform evaluating performance under delayed communication conditions. The digital twin architecture allows for predictive-assisted operation, with forward-state estimation helping to address the communication gaps. This approach enhances operator awareness, reduces the likelihood of collisions, and improves mission reliability.

There are substantial benefits to addressing this problem. This system improves rover safety, extends hardware longevity, and improves operational efficiency. Operationally, it reduces the financial and logistical risks associated with hardware loss and mission delays. Functionally, the project delivers to the Florida Space Institute a validated prototype system and documented methodology that can serve as a foundation for continued research. Risks and challenges with this project come from challenges to accuracy of the digital twin. This can be

affected by factors such as sensor fidelity and system calibration, and inaccurate predictions could create controller overreliance on the virtual model. This would compound navigational errors, rather than mitigate them.

Despite these challenges, the expected value of this work outweighs its risks. The project establishes a structured framework for studying latency-tolerant teleoperation while fostering workforce development in space systems engineering. By demonstrating the effect of predictive modeling and digital twin technologies on mitigation of latency constraints, this work contributes practical solutions to benefit the field of intelligent remote systems.

14. Acknowledgments

The team gives their thanks to all persons who have helped in preparing this proposal, and in the formation of this project:

Dr. Luis Rabelo

Dr. Mansoor Mollaghasemi

FSI Representative, Christian Vincent

Project Support, Carson Gray

15. References

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Appendices

Project Charter



Department of Industrial Engineering and Management Systems

College of Engineering and Computer Science

University of Central Florida

Industrial Engineering Capstone Senior Design/Praxis of Data Science Course

SENIOR DESIGN PROJECT CHARTER

Project Sponsor:	<i>Florida Space Institute</i>
Project Manager:	<i>Jennifer Cifuentes</i>
Co – Project Manager	
E-mail Address:	<i>jennifer.cifuentes@ucf.edu</i>
Phone Number:	<i>+1 (321)900-2878</i>

Process Impacted:	<i>Enter the name of the impacted process(es) here</i>		
Expected Start Date:	<i>Remote rover teleoperation and state monitoring under simulated lunar communication delay conditions</i>		
Expected Completion Date:	<i>April 19, 2026</i>		
Green Belts Assigned:	<i>Luis Rabelo, Ph.D.</i>	Black Belts Assigned:	<i>Ahmed Mohammed Ph.D.</i>
Mentor:	<i>Mansoorah Mollaghasemi, Ph.D.</i>	Master Black Belt Assigned:	<i>Jose Flores, Ph.D.</i>

Element	Description of Element	Project Charter
1. Title of Project	What is the title of the improvement project?	RASSOR Rover Remote Operation and AI-Driven Digital Twin Development
2. Project Description	Brief background and project's purpose	To establish methods of addressing time delay in lunar communications.
3. Process Description	Brief description of the process in which opportunity exists	Remote rover operation under simulated lunar conditions is impacted by communication latency and limited environmental feedback. Operators experience delays between issued commands and rover response, reducing efficiency and increasing operational risk. An opportunity exists to integrate a predictive digital twin system to improve synchronization, situational awareness, and navigation performance.
4. Business Case	Brief explanation of the expected financial improvement, or other justification	Expected improvements include reduction of risk of rover damages, improvements to mission time duration, and efficiency of rover operation.

5. Benefit to External Customers	Who are the final customers and what benefits will they see?	FSI will receive a validated prototype system and documented methodology that can serve as a foundation for continued research in addressing communication latency.		
6. Problem Statement	Problem and goal statement (project's purpose)	Current remote rover operations suffer from communication latency, limited environmental awareness, and lack of synchronized feedback systems, reducing operational efficiency and rover safety during remote navigation. To tackle in-field communication latency by allowing the digital twin to predict the rover's current state, improving navigation, operator awareness, and overall mission performance.		
7. Project Objectives	What are the performance metrics such as throughput, quality, inventory, defects, yield, costs, etc.? What improvement is targeted?	Performance Metric	Current (As-Is) Value	Future (To-Be) Value
		Operator Feedback During Latency	No predictive state feedback	Predictive digital twin visual
		Positional Awareness	Lack of data collection	Integrated sensor-based state modeling
		Navigation Efficiency	Manual drive only	Design predictive modeling architecture
		System Integration	No autonomous	Integrated rover with digital twin system
8. Scope Statement	Which parts of the process will be investigated and which parts will be excluded? What are the major tasks to complete?	The team will construct one functional RASSOR rover and set up for internet based teleoperation control of the rover. The team will not develop systems in actual lunar or space environments or utilize space qualified hardware and systems		
9. Proposed Methods	Which design and analysis concepts, techniques and tools are going to be used?	The team will use systems engineering principles, iterative prototyping, and structured testing to develop and validate a physics-based digital twin with steps towards AI-assisted state prediction. Performance testing and system integration will support evaluation.		
10. Key Deliverables	What are the tangible and/or intangible objects that are produced as a result of the project and that relate	A completed, drivable rover - The final rover built and operating, able to be driven, and outfitted with sensors for positional data acquisition.		

	directly to and align with the objectives of the project?	Digital Twin System - A physics based Digital Twin of the physical rover that can model the effects of motion commands on the rover. Architecture for AI assisted motion control - Setup for autonomous AI rover navigation, to be built upon in the second semester of the project. Handoff Documentation - Well organized documentation of the entire process of design and control, to allow the second semester to have the necessary information to take over the project.	
11. Schedule	Give the key project milestones and dates	Project Start Date:	1/16/26
	“D” – Define Phase	“D” Completion Date:	2/21/26
	“M” – Measure Phase	“M” Completion Date:	3/28/26
	“A” – Analysis Phase	“A” Completion Date:	3/31/26
	“I” – Improve Phase	“I” Completion Date:	4/8/26
	“C” – Control Phase	“C” Completion Date:	4/11/26
12. Team Members	Names of the project team members (in alphabetical order of last names)?	Jennifer Cifuentes, Noah Choi, Sean Crites, Matteo D’Agostino, Mattias Escudero, Jonathan Gyure, Abdiel Joseph	
13. Costs and Resources	What are the estimated project costs and needed resources?	The estimated project cost will be \$1461.55.	

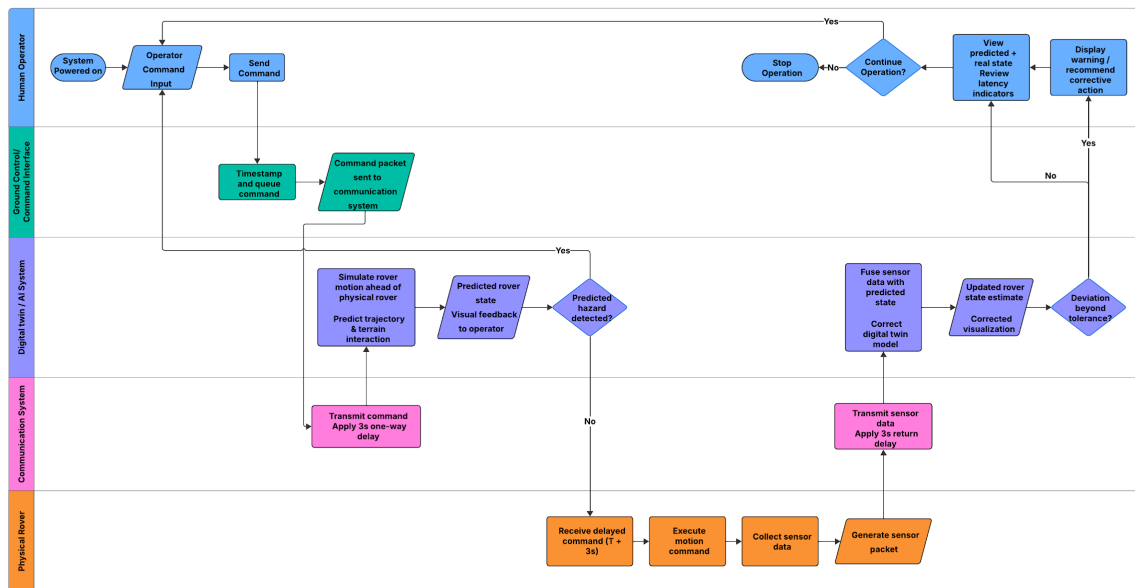


Figure 4: Command Process Map

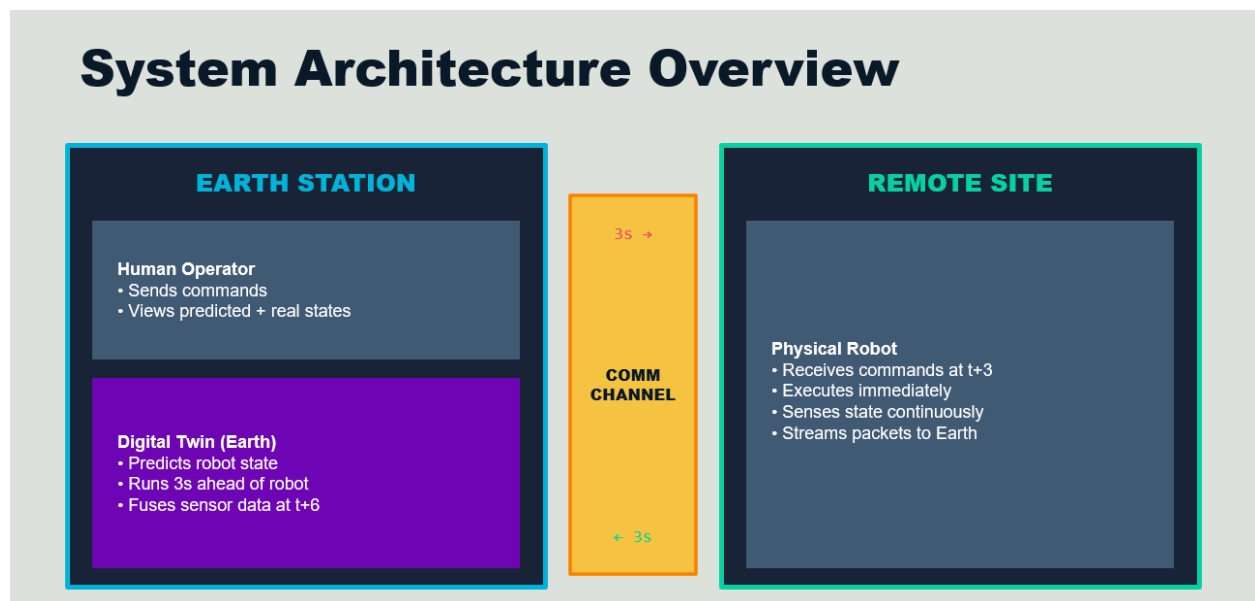


Figure 5: Overview of Command System

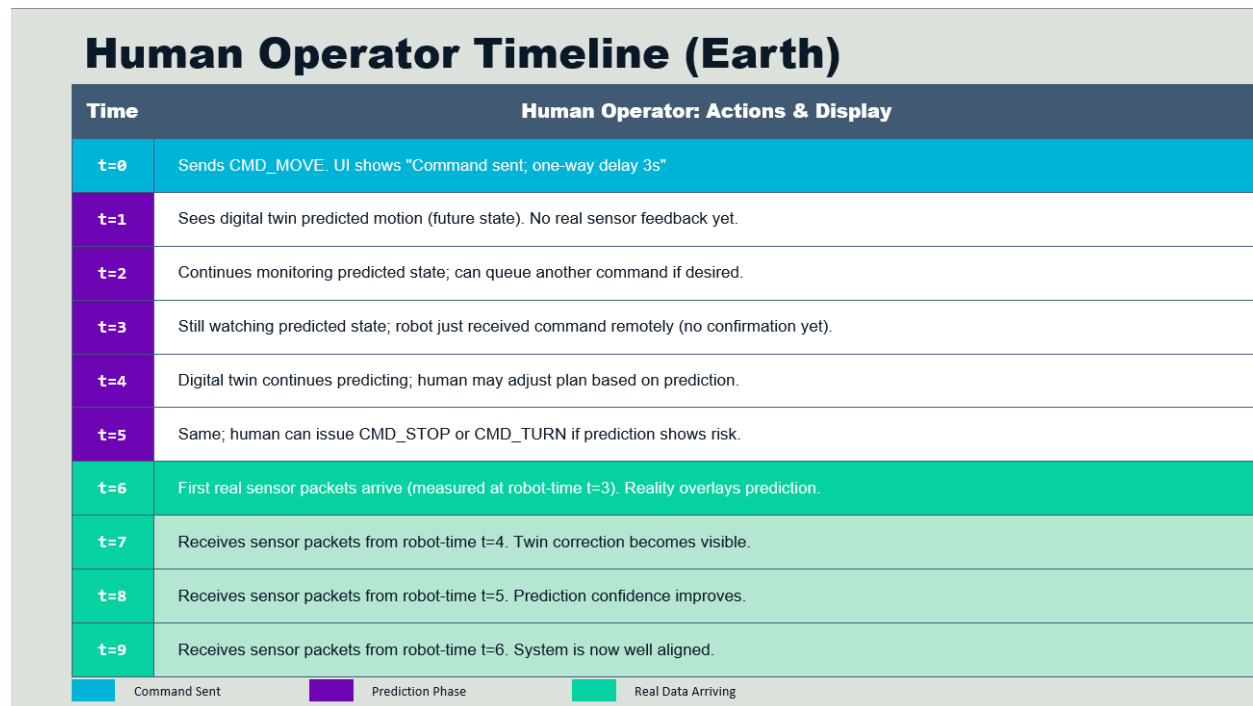


Figure 6: 9 second Timeline, as viewed by Operator

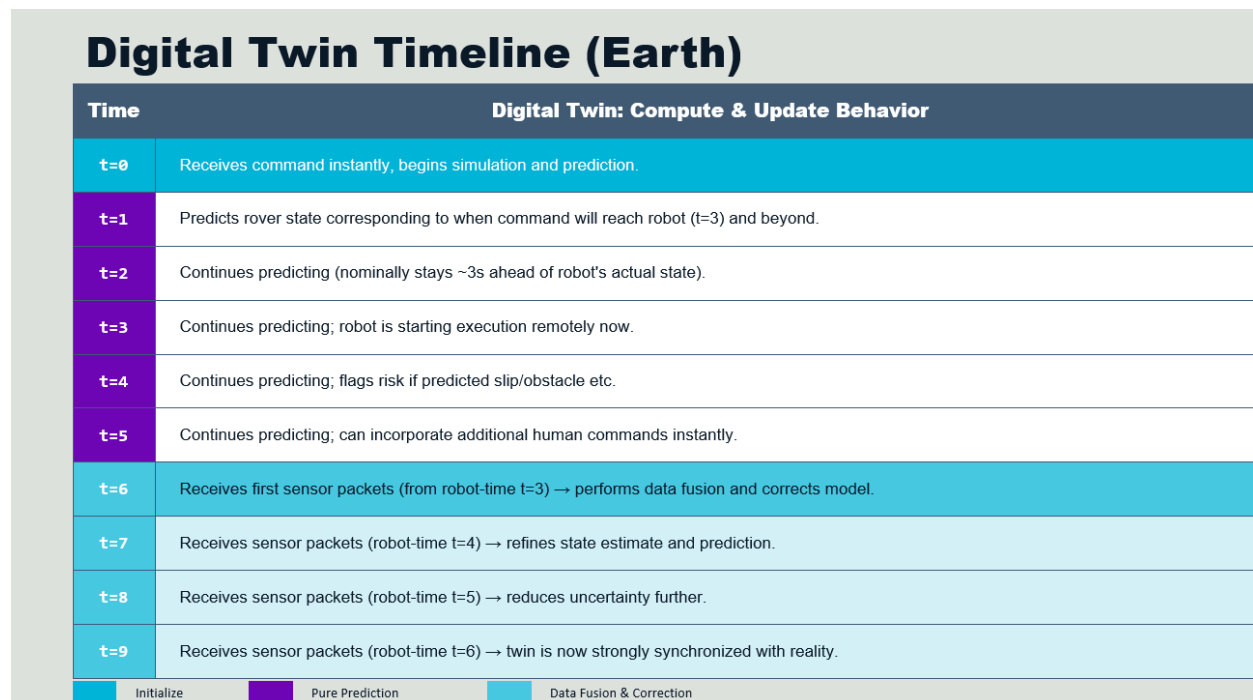


Figure 7: 9 second Timeline, as viewed by Digital Twin

Physical Robot Timeline (Remote Site)			
Time	Execution State	Sensing State	Sending State
t=0	Idle (no command yet)	Sensors may be on (heartbeat)	No command-related telemetry
t=1	Idle	Same	Same
t=2	Idle	Same	Same
t=3	CMD arrives → Execution starts!	Measures state at t=3	Sends S(3) → arrives Earth t=6
t=4	Executing motion	Measures state at t=4	Sends S(4) → arrives Earth t=7
t=5	Executing motion	Measures state at t=5	Sends S(5) → arrives Earth t=8
t=6	Executing motion	Measures state at t=6	Sends S(6) → arrives Earth t=9
t=7	Executing motion	Measures state at t=7	Sends S(7) → arrives Earth t=10
t=8	Executing motion	Measures state at t=8	Sends S(8) → arrives Earth t=11
t=9	Execution finishing	Measures state at t=9	Sends S(9) → arrives Earth t=12

S(n) = Sensor packet measured at robot-time n, sent immediately, arrives at Earth at time n+3

Figure 8: 9 second Timeline, as viewed by physical robot

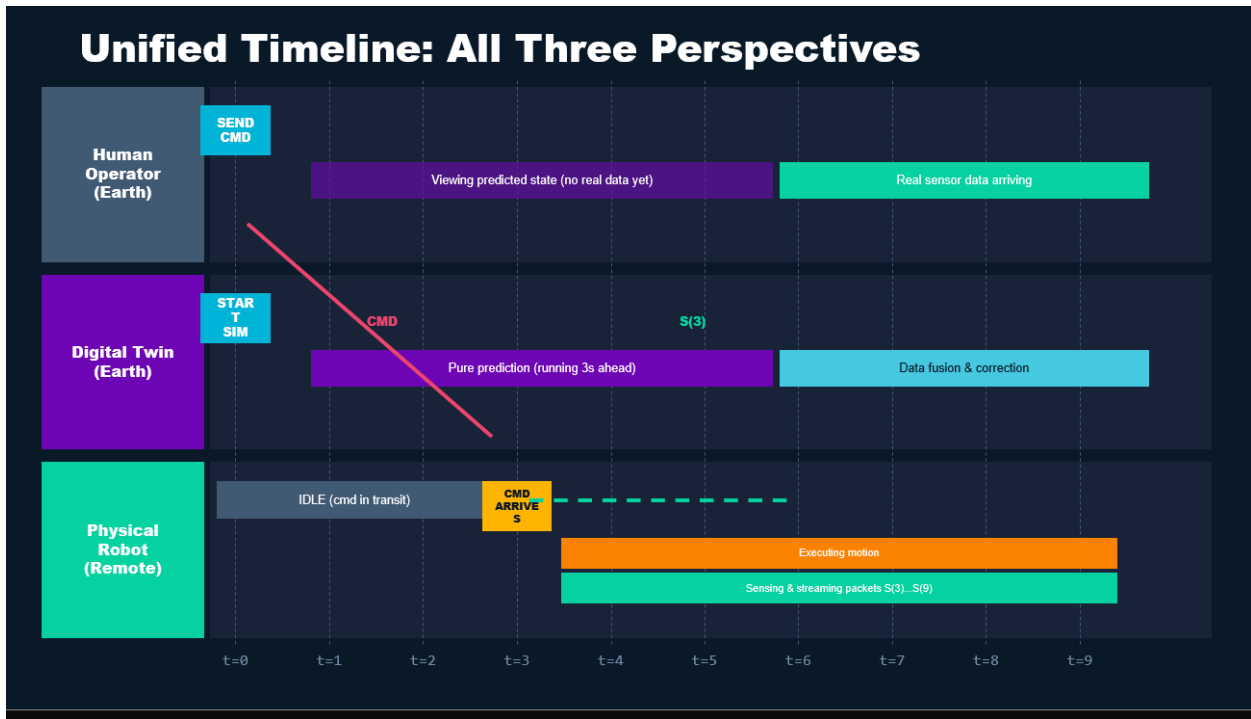


Figure 9: 9 second timeline combined view